CSE225 Data Structures Fall 2011 7/10/2011 Student Name:_____
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Q1. (10+15 = 25 pts)
Find the execution time ***t(n)*** **and** the asymptotic upper bound (***O(.)***) of the following pieces of code assuming *n* is the input size:

i.
```
for (i=0; i<=n; i++)
  for (j=1; j<=i; j*=2)
    statement block;
```

$$t(n)=0+1+2+2+3+3+3+3+4+\ldots.+(\lfloor \log_2 n \rfloor+1)=\sum_{i=1}^{n}(\lfloor \log_2 i \rfloor+1)$$

$$t(n)\in O(n\log_2 n)$$

ii.
```
for (i=1; i<=n; i*=3)
  for (j=1; j<=n; j*=2)
    statement block;
```

$$t(n)=(\lfloor \log_2 n \rfloor+1)*(\lfloor \log_3 n \rfloor+1)$$

$$t(n)\in O(\log_2^2 n)$$

Q2. (5+5+5= 15 pts)
Find the asymptotic upper bound (***O(.)***) of the following run times **in terms of** the input size ***n***.

i. $t(n)=\sum_{i=1}^{n} i\sum_{j=1}^{i} j=\sum_{i=1}^{n}\frac{i^3+i^2}{2}=\frac{1}{2}(\frac{1}{4}(n^4+2n^3+n^2))+\frac{1}{2}(\frac{1}{6}(2n^3+3n^2+n))$ $t(n)\in O(n^4)$

ii. $t(n)=\prod_{i=1}^{\lfloor \frac{n}{2} \rfloor} i=\lfloor \frac{n}{2} \rfloor !$ $t(n)\in O(\frac{n}{2}!)$

iii. $t(n)=\sum_{i=1}^{n-1} a^i=\frac{1-a^n}{1-a};$ $a \in N$ $t(n)\in O(a^n)$

Q3. (4+3+3=10 pts)
At each choice are given the asymptotic upper bounds of the run times of algorithms that solve the same problem. Put the algorithms in order starting from best towards worst per problem!

i. $O(n^2)$ (3) $O(n\log n)$ (1) $O(n\log\{\log(n^n)\})$ (2/1*) *second and third may be regarded same or second better than third

ii. $O(\log_{10} n)$(1) $O(\log_2 n)$ (3) $O(\log_5 n)$ (2)

iii. $O(n!)$ (1) $O(n^n)$ (3) $O((n+1)^{(n-1)})$ (2)

Q4. (4+3+3=10 pts)
Specify the index k of the array in terms of the indices i,j of the following square matrix m_{nxn} such that you optimize the use of memory space! That is, find f such that $k=f(i,j)$

$$\begin{bmatrix} 4 & -1 & 6 & \cdots & -2 \\ & & & 5 & \\ & & 11 & & \\ & \iddots & & & \\ 7 & 8 & 12 & \cdots & 9 \end{bmatrix}$$

Hint: Use a piecewise function

$$k=\begin{cases} j & ; i=0 \\ n+i-1 & ; 0<i<n-1 \\ 2n-2+j & ; i=n-1 \end{cases}$$

(indices starting from 0)

OR

$$k=\begin{cases} j & ; i=1 \\ n+i & ; 1<i<n \\ 2n-1+j & ; i=n \end{cases}$$

(indices starting from 1)

Q5. (4+3+3=10 pts)
Would it be possible to come up with a better optimization on memory space for the following matrix?

$$\begin{bmatrix} 2 & 4 & 6 & \cdots & 2n \\ & & & 2n+1 & \\ & & 2n+2 & & \\ & \iddots & & & \\ 3n-1 & 3n+1 & 3n+3 & \cdots & 5n-3 \end{bmatrix}$$

Hint: Note that now it is not only the places in the matrix that follow a specific order but also the the entries (i.e., the data) themselves at those places.

$$k=\begin{cases} 2(j+1) & ; i=0 \\ 2n+i & ; 0<i<n-1 \\ 3n-1+2j & ; i=n-1 \end{cases}$$

(indices starting from 0)

OR

$$k=\begin{cases} 2j & ; i=1 \\ 2n+i-1 & ; 1<i<n \\ 3n+2j-3 & ; i=n \end{cases}$$

(indices starting from 1)